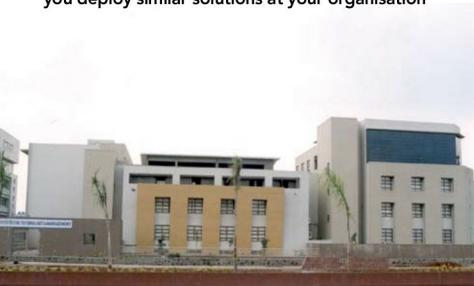




It's Open Source All the Way for ITM

ITM, one of the leading technology and management institutes in India, could be dubbed a role model for other such institutes. This fascinating case study will give you an idea about the FOSS-based IT infrastructure of the institute, and also help you deploy similar solutions at your organisation



ITM, Navi Mumbai

Swapnil Bhartiya

The ITM Group of Business Schools grew from one business school in Mumbai to seven campuses by the year 2004. Starting with a two-year, full-time masters programme in 1991, this group currently conducts 15 programmes covering financial markets, retail, pharmaceuticals, healthcare, risk management, human resources, etc.

IT infrastructure

Being an educational institute, the IT needs of ITM were a bit different and more demanding than an ordinary organisation. There were typically three kinds of users—the staff, faculty and students/visitors.

The hardware needed by the institute required to focus on providing wired desktops for staff and faculty members, while the students' needs were met using

Wi-Fi connectivity for their laptops.

The core computing requirement of staff members was office applications, e-mail and Web-based ERP, while the faculty needed additional access to a variety of online resources (EBSCO, PROQUEST and HBR online) and in-house databases (Prowess and Capital Line). An intranet is used for disseminating local information.

The institution uses a mix of branded and assembled servers and client machines. And it uses the laptops of most of the well-known brands such as Toshiba, IBM, HP, Fujitsu and Acer.

The network infrastructure is completely powered by open source software, and all staff and faculty desktops too use open source applications running on different versions of Fedora. Volume licences of Windows XP Professional and MS Office 2007 Professional are being used for laptops. However, all other pre-loaded applications on laptops are open source.

The institute uses a mix of thin clients and standalone machines. There are 89 thin clients for the staff and faculty—a few of which are thin clients-cum-print servers. The number of laptops available for students is over 400.

Connecting people: networking security

The campus is divided into two independent networks. The wired network is used mainly by staff and faculty, while the wireless network is dedicated for students. Only the data exchange server, the Prowess database and intranet server have data common to the faculty and students. In each of these servers, the institute has two network cards that provide connectivity to each group while keeping the networks independent. All the servers—including the DNS, DHCP, LDAP, the Web server (intranet), LTSPs, NFS, the data exchange server and the NAS box (back-up server)—are in the staff network, which is completely isolated from the students.

To provide the much needed and efficient connectivity among the servers and clients, structured cabling is in place, connecting two server rooms located at diagonally opposite corners of the campus. The server rooms house the switch racks, with cables running directly to each machine on the campus or one of the many Wi-Fi access points. There are 18 access points by NetGear with RADIUS authentication deployed to meet all requirements.

Security

This kind of widespread network needs a lot of security measures—some physical and some at the virtual level. Besides physical security, the institute ensures the normal